

Master Specialization Proposal: Comprehensive Climate, Glacial, and Watershed Analysis

This proposal details a massive, unified academic implementation plan merging the objectives of Climate Change, Glacial Retreat, and Watershed Modeling analysis. It leverages **ArcGIS Pro (ArcHydro), ENVI, RStudio, and pure Python environments** to create a robust analytical pipeline.

Furthermore, to elevate the project to an enterprise-grade Geo-AI architecture, the specialization progresses into advanced topics including **Agentic Orchestration (LangGraph), Multi-Task PyTorch, Multi-Temporal Change Detection, PostGIS, and Full-Stack web development**.

The learning process is divided into 11 sequential, academic **Chapters**, ensuring a step-by-step understanding of how to build and scale these complex tools.

Summary of Resources and Techniques

Data Resources:

- **Optical & Thermal Sensors:** Landsat, Sentinel-2, MODIS.
- **Radar & Elevation:** Sentinel-1 SAR, SRTM / ALOS PALSAR DEMs.
- **Climatic Datasets:** WorldClim, CHIRPS.
- **In-Situ Data:** Local meteorological weather stations (e.g., CR2 Chile).

Advanced Analytical Techniques:

- **Agentic AI:** LangGraph for autonomous geospatial orchestration.
- **Deep Learning (PyTorch):** Multi-task CNNs for simultaneous lake, snow, and change detection segmentation.
- **Multi-Temporal Change Detection:** Time-series analysis across decades using fused SAR and Optical data.
- **Hydrological Modeling (ArcHydro / PySheds):** Watershed delineation and water balance.
- **Spatial Databases:** PostgreSQL + PostGIS for scalable vector storage.
- **Full-Stack Web GIS:** FastAPI + React + MapLibre GL JS for interactive dashboards.

Selected Region of Interest (ROI)

Torres del Paine National Park & Punta Arenas (Grey River Basin), Magallanes Region, Chile.

Detailed Project Structure and Academic Flow

CHAPTER 1: Climatic Variables & Image Processing

Academic Objective: Foundation in data acquisition, atmospheric correction, and climate modeling.

- **01_stac_multisensor_download.py**: (Concept: Programmatic acquisition). Uses Copernicus API and AWS STAC to download multi-sensor data.
 - **02_atmospheric_correction.py**: (Concept: TOA to BOA reflectance physics). Radiometric/Atmospheric correction using Python (**rasterio/Py6S**) and ENVI.
 - **03_station_ml_interpolation.py**: (Concept: ML for missing data). Training Random Forest on weather station data to predict continuous climate surfaces.
 - **04_precipitation_dual_analysis**: (Concept: R vs Python spatial analysis). CHIRPS historical anomalies and ArcGIS IsoCluster climatic zone classification.
 - **05_uhi_modis_mapping.py**: (Concept: Urban microclimates). Urban Heat Island mapping over Punta Arenas using thermal data.
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CHAPTER 2: Glacial Retreat, SAR Dynamics, and AI

Academic Objective: Tracking ice loss using multi-sensor approaches (Radar/Optical) and foundational Deep Learning.

Chapter_02/

- **01_glacier_area_perimeter.py**: (Concept: Vector GIS change detection). ArcGIS Pro/Python script to calculate historical Grey Glacier retreat.
 - **02_sentinel1_sar_analysis.py**: (Concept: Cloud-penetrating radar physics). Classify dry snow vs. wet snow vs. ice using SAR backscatter.
 - **03_xarray_ndwi_multitemporal.py**: (Concept: Spectral indices). Mapping Grey Lake expansion across decades.
 - **04_pytorch_lake_segmentation.py**: (Concept: Semantic segmentation). U-Net CNN for automated glacial lake extraction.
 - **05_arcpy_ndsi_snow_cover.py**: (Concept: Snow tracking). Automated ArcGIS Pro script for NDSI mapping.
 - **06_multitemporal_change_detection.py**: (Concept: Multi-temporal sensor fusion). Fusing Sentinel-1 SAR and Sentinel-2 Optical time-series data to algorithmically detect land-cover changes and glacial calving events between two dates (T1 vs T2).
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CHAPTER 3: Hydrology and Watershed Modeling

Academic Objective: Understanding the mechanics of water flow, terrain modeling, and basin management.

Chapter_03/

- **01_dem_hydro_conditioning.py**: (Concept: Terrain correction). Filling sinks and applying TIN corrections to DEMs.
- **02_archydro_basin_delineation.py**: (Concept: Automated flow routing). Using Archydro and PySheds to delineate the Grey River Watershed.
- **03_morphometric_parameters.py**: (Concept: Basin physical traits). Calculate form factor, mean slope, and plot the Hypsometric Curve.
- **04_drainage_and_strahler.py**: (Concept: Stream hierarchy). Calculating drainage density and Strahler stream order.

- **05_water_balance_isohyets.py**: (Concept: Spatial hydrology inputs/outputs). Generate Isohyets and Isotherms for a spatial Water Balance model.
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CHAPTER 4: Vulnerability and Ecosystem Impacts

Academic Objective: Understanding the biological and human consequences of the physical changes.

Chapter_04/

- **01_maxent_modeling (R/Python)**: (Concept: Species Distribution Modeling). Map how flora/fauna will migrate as the watershed's climate shifts.
 - **02_vulnerability_index_mce.py**: (Concept: Multi-Criteria Evaluation). Combine precipitation, glacial retreat, and flood risks into a "Climate Vulnerability Heatmap".
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CHAPTER 5: Cartography and Management Plans

Academic Objective: Communicating spatial data clearly via standard GIS outputs.

Chapter_05/

- **01_map_automation_layout.py**: (Concept: Cartographic automation). **arcpy.mp** script generating standardized PDF maps.
 - **02_watershed_management_report.py**: (Concept: Reporting). Compile stats, water balance, and vulnerability indices into a final text/PDF management plan.
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CHAPTER 6: Capstone Project - The Linear Unified Pipeline

Academic Objective: Synthesize previous modules into a single, automated codebase.

Chapter_06/

- **main_linear_pipeline.py**: (Concept: System integration). A massive Python script that linearly triggers downloads, preprocessing, AI inference, watershed delineation, and report generation in one continuous execution loop.
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ADVANCED ARCHITECTURES

CHAPTER 7: Agentic Orchestration (LangGraph)

Academic Objective: Move from rigid, linear scripting to autonomous, state-machine driven AI agents.

Chapter_07/

- **Concept:** Instead of a script running top-to-bottom, we build specialized AI agents that "decide" when to act based on the data state.

- **Action:** Build a LangGraph network with an **AcquisitionAgent**, **GlacierAgent**, **HydrologyAgent**, and **ReportAgent**. They autonomously pass data payloads (like GeoTIFFs) between each other, handling errors and retries intelligently.

CHAPTER 8: The Cascade Effect Modeling

Academic Objective: Move from isolated metrics to modeling a physical chain-reaction.

Chapter_08/

- **Concept:** Disasters don't happen in isolation. We will programmatically link the models to prove cause-and-effect.
- **Action:** A Python framework that specifically tracks the statistical causality: *Temperature Anomaly (Ch1)* → *Lake Expansion (Ch2)* → *Altered Water Balance (Ch3)* → *High Vulnerability Niche Shift (Ch4)*.

CHAPTER 9: Multi-Task PyTorch Model

Academic Objective: Upgrade standard Deep Learning to advanced multi-head architectures.

Chapter_09/

- **Concept:** Running multiple separate neural networks is computationally expensive. Multi-task learning predicts several things at once.
- **Action:** Upgrade the U-Net from Chapter 2 into a **Multi-Head CNN**. A single PyTorch model that ingests Optical, SAR, and DEM tensors simultaneously to output masks in one forward pass:
 - (1) Glacial Lake Segmentation
 - (2) Snow/Ice Cover
 - (3) Flood Risk Zones
 - **(4) Multi-Temporal Change Detection Head:** Anomaly scoring that highlights structural changes in the terrain between two different years.

CHAPTER 10: Spatial Database Integration (PostGIS)

Academic Objective: Move from flat files (Shapefiles/GeoTIFFs) to enterprise spatial databases.

Chapter_10/

- **Concept:** Flat files are hard to query over time. Databases allow for massive spatial SQL queries.
- **Action:** Set up a PostgreSQL + PostGIS database via Docker. Write Python (**SQLAlchemy**, **GeoAlchemy2**) scripts to ingest the output polygons (lakes, watersheds, vulnerability zones) and raster metadata directly into the database.

CHAPTER 11: Interactive Dashboard (Full-Stack GIS)

Academic Objective: Move from static PDF maps to real-time, interactive Web GIS.

Chapter_11/

- **Concept:** Stakeholders need interactive platforms, not just PDFs.

- **Action:** Build a **FastAPI** backend that queries the PostGIS database, and a **React + MapLibre GL JS** frontend to visualize the Climate Cascade, Glacial Retreat, and Watershed changes dynamically over a web browser.